

MAT1002 Final Examination

Tuesday, May 13th, 2025

Time: 13:30 PM — 16:30 PM

Notes and Instructions

1. *No books, no notes, no dictionaries, and no calculators.*
2. *The maximum score of this examination is **132**.*
3. *There are **13** questions (with parts) in total.*
4. *The symbol $[N]$ at the beginning of a question indicates that the question is worth N points.*
5. *Answer all questions on the **answer book**.*
6. *Show your intermediate steps **except Question 1** — answers without intermediate steps will receive minimal (or even no) marks.*
7. *Express irrational numbers in exact forms instead of decimal forms; e.g., write $\sqrt{2}$ instead of $1.414\dots$, and write $\ln 2$ instead of $0.693\dots$*

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MAT1002 Final Exam Questions

1. [27] Short questions: no intermediate step is required.

- (i) True (T) or False (F): The 3D space without the origin is a simply connected region.
- (ii) True (T) or False (F): Suppose the function $z = f(x, y)$ is continuous with respect to x for any fixed y , and continuous with respect to y for any fixed x , then it is a continuous function.
- (iii) True (T) or False (F): The vector field $\mathbf{F} = (\frac{x}{x^2+y^2}, \frac{y}{x^2+y^2})$ passes the component test, thus it is conservative in its domain.
- (iv) True (T) or False (F): Suppose $z = f(x, y)$ is continuous function on the region R where $R = R_1 \cup R_2$. Then we have

$$\iint_R f(x, y) \, dA = \iint_{R_1} f(x, y) \, dA + \iint_{R_2} f(x, y) \, dA.$$

- (v) Rewrite the following integral in *Cylindrical coordinates*, in the order of $dzd\theta dr$:

$$\int_{-1}^1 \int_0^{\sqrt{1-x^2}} \int_0^{x^2+y^2} f(x, y, z) \, dz \, dy \, dx.$$

- (vi) Find the direction derivative for the function $w = xy + \cos(z) - \frac{y}{x}$ at the point $(2, 1, 0)$ in the direction $(\frac{1}{3}, \frac{2}{3}, \frac{2}{3})$.
- (vii) Find the direction (the unit vector) that parallel to both the plane $3x - 2y + z = 0$ and $x + y - 3z = 0$.
- (viii) Find the quadratic approximation at $(0, 0)$ of $z = \ln(1+x) \sin(y)$.

- (ix) Compute $\sum_{n=1}^{\infty} \frac{2n+1}{n!} = (\quad)$.

2. [15] Determine the following limits. If the limit exists (as a real number), find the value and show your steps; otherwise, explain why it does not exist.

- (i) $\lim_{(x,y) \rightarrow (0,0)} \frac{\sin(xy)}{x^2+y^2}$

(ii) $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^4 + y^2}$

(iii) $\lim_{(r,\theta) \rightarrow (0,\pi/2)} \frac{r \cos \theta}{r^2 + 1}$

3. [6] If $f(x, y) = x \sin y + y \ln x$, find the second-order derivatives: $\frac{\partial^2 f}{\partial x^2}$, $\frac{\partial^2 f}{\partial y \partial x}$, $\frac{\partial^2 f}{\partial y^2}$, $\frac{\partial^2 f}{\partial x \partial y}$.

4. [6] Let Surface S be characterized by the equation $z = x^2 - y^2$. Find the tangent plane M and normal line L to the surface at point $(2, 1, 3)$.

5. [6] Find the point on the intersection line between plane $2x + y - z = 4$ and plane $x + y + z = 3$ that has minimum distance from the origin $(0, 0, 0)$

6. [8] Find and classify the critical points (as local maxima, local minima, or saddle points) of the function: $f(x, y) = x^3 + y^3 - 3xy$.

7. [6+6] Evaluate the following integrals.

(a) $\iint_R (3 - x - y) dA$, where R is the region enclosed by the parabola $y = x^2$ and the line $y = x + 2$.

(b) $\int_0^3 \int_{\sqrt{x/3}}^1 e^{y^3} dy dx$

8. [4+4] Consider a solid bounded from below by a sphere of radius one centered at the origin and from above by the paraboloid $z = 1 - x^2 - y^2$.

(a) Express the volume of the solid in terms of a triple integral in cylindrical coordinates.

(b) Find the volume of the solid by evaluating the integral in (a).

9. [3+5] Consider the integral

$$\int_0^{2/3} \int_y^{2(1-y)} (x + 2y) e^{y-x} dx dy.$$

(a) Sketch the integration region R in the xy -plane.

- (b) Using the transformation $u = x + 2y$, $v = x - y$, express it as an integral over a suitable region G in the uv -plane, and sketch the region G (**DO NOT** evaluate the integral.)

10. [4+4+4] (a) For what values of b and c will

$$\mathbf{F} = (y^2 + 2cxz)\mathbf{i} + y(bx + cz)\mathbf{j} + (y^2 + cx^2)\mathbf{k}$$

be a gradient field?

- (b) Find a potential function of

$$\mathbf{F} = (y^2 + 4xz)\mathbf{i} + (2xy + 2yz)\mathbf{j} + (y^2 + 2x^2)\mathbf{k}$$

- (c) If $\mathbf{F}$ in (b) is a force field, find the work done by $\mathbf{F}$ on the particle that moves from the point $(0, 0, 0)$ to $(1, 2, 2)$ along the line segment joining them.

11. [8] (a) Find the flux of the fields $\mathbf{F}(x, y) = \langle x, y \rangle$ across the closed path consisting of:

1. The right semicircle from $(0, 2)$ to $(0, -2)$
2. The line segment on the y -axis from $(0, -2)$ to $(0, 2)$

- (b) Find the circulation of the fields $\mathbf{F}(x, y) = \langle -y, x \rangle$ around the closed path consisting of:

1. The right semicircle from $(0, 2)$ to $(0, -2)$. (clockwise)
2. The line segment on the y -axis from $(0, -2)$ to $(0, 2)$

12. [8] Consider the vector field $\mathbf{F}(x, y, z) = \langle x^2y, y^2z, z^2x \rangle$. Compute the outward flux of $\mathbf{F}$ through the surface of the cube bounded by the planes $x = 0$, $x = 1$, $y = 0$, $y = 1$, $z = 0$, and $z = 1$.

13. [8] Consider the vector field $\mathbf{F}(x, y, z) = \langle y^2, z^2, x^2 \rangle$. Let S be the part of the paraboloid $z = 4 - x^2 - y^2$ that lies above the xy -plane, oriented upward. The boundary of S is the curve C (a circle in the xy -plane). Compute the surface integral

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} d\sigma$$

using Stokes' Theorem.

Q1 Answer:

(i) T.

(ii) F.

(iii) F.

(iv) F.

(v) If the students miss the r in the integrand, 1 point off.

$$\int_0^1 \int_0^\pi \int_0^{r^2} f(r \cos \theta, r \sin \theta, z) r \, dz d\theta dr$$

(vi) $\frac{17}{12}$.

(vii) $\frac{(1,2,1)}{\sqrt{6}}$. It is OK to write $\frac{(-1,-2,-1)}{\sqrt{6}}$.

(viii) xy .

(ix) $3e - 1$.

Q2 Answer:

- (i) The limit does not exist (2 points).

Approaching along $y = x$ gives:

$$\lim_{x \rightarrow 0} \frac{\sin(x^2)}{2x^2} = \frac{1}{2}. \quad (2 \text{ points})$$

Approaching along $y = 0$ gives:

$$\lim_{x \rightarrow 0} \frac{0}{x^2} = 0. \quad (1 \text{ points})$$

Since the limits differ, the limit does not exist.

- (ii) The limit does not exist (2 points).

Approaching along $y = kx^2$ gives:

$$\lim_{x \rightarrow 0} \frac{kx^4}{x^4 + k^2x^4} = \frac{k}{1 + k^2}, \quad (3 \text{ points})$$

which depends on k .

Thus, the limit does not exist.

- (iii) The limit is 0 (2 points).

Substituting $r \rightarrow 0$ and $\theta \rightarrow \pi/2$:

$$\lim_{(r, \theta) \rightarrow (0, \pi/2)} \frac{r \cdot 0}{0 + 1} = 0. \quad (3 \text{ points})$$

Q3 Answer:

First-order partial derivatives:

$$\frac{\partial f}{\partial x} = \sin y + \frac{y}{x}, \quad \frac{\partial f}{\partial y} = x \cos y + \ln x. \text{(2 points)}$$

Second-order partial derivatives:

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} &= -\frac{y}{x^2}, \\ \frac{\partial^2 f}{\partial y \partial x} &= \cos y + \frac{1}{x} = \frac{\partial^2 f}{\partial x \partial y}, \\ \frac{\partial^2 f}{\partial y^2} &= -x \sin y. \text{(4 points)} \end{aligned}$$

Q4 Answer:

The tangent plane to the surface $z = f(x, y) = x^2 - y^2$ at the point $(x_0, y_0, z_0) = (2, 1, 3)$ is given by: $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$, where f_x and f_y are the partial derivatives of f with respect to x and y , respectively.

Compute the partial derivatives: $f_x = 2x$, $f_y = -2y$. Evaluate at $(2, 1)$: $f_x(2, 1) = 4$, $f_y(2, 1) = -2$.

Thus, the equation of the tangent plane M is: $z - 3 = 4(x - 2) - 2(y - 1)$. Simplify: $z = 4x - 8 - 2y + 2 + 3 = 4x - 2y - 3$. So, the tangent plane is: $M : 4x - 2y - z - 3 = 0$. (3 points)

The normal line L to the surface at $(2, 1, 3)$ has direction vector given by the gradient of f at $(2, 1)$: $\nabla f(2, 1) = (4, -2, -1)$. The parametric equations

of the normal line are: $L : \begin{cases} x = 2 + 4t, \\ y = 1 - 2t, \\ z = 3 - t. \end{cases}$ (3 points)

Q5 Answer:

First, find the intersection line of the two planes. Solve the system:

$$\begin{cases} 2x + y - z = 4, \\ x + y + z = 3. \end{cases}$$

Add the two equations to eliminate z : $3x + 2y = 7 \implies y = \frac{7-3x}{2}$.

Substitute y into the second equation to solve for z : $x + \frac{7-3x}{2} + z = 3 \implies z = 3 - x - \frac{7-3x}{2} = \frac{x-1}{2}$.

The parametric equations of the intersection line are: $\begin{cases} x = t, \\ y = \frac{7-3t}{2}, \\ z = \frac{t-1}{2}. \end{cases}$ (3 points)

The distance D from the origin $(0, 0, 0)$ to a point (x, y, z) on the line is:

$$D = \sqrt{x^2 + y^2 + z^2} = \sqrt{t^2 + \left(\frac{7-3t}{2}\right)^2 + \left(\frac{t-1}{2}\right)^2}.$$

To minimize D , minimize D^2 : $D^2 = t^2 + \left(\frac{7-3t}{2}\right)^2 + \left(\frac{t-1}{2}\right)^2 = \frac{7t^2}{2} - 11t + \frac{25}{2}$.

Take the derivative with respect to t and set it to zero: $\frac{dD^2}{dt} = 7t - 11 = 0 \implies t = \frac{11}{7}$.

Substitute $t = \frac{11}{7}$ back into the parametric equations to find the point:

$$x = \frac{11}{7}, \quad y = \frac{8}{7}, \quad z = \frac{2}{7}.$$

Thus, the point is: $\left(\frac{11}{7}, \frac{8}{7}, \frac{2}{7}\right)$. (3 points)

Or

$$\text{line: } \begin{cases} x = 2t + 1, \\ y = 2 - 3t, \\ z = t. \end{cases}$$

$$D^2 = 14t^2 - 8t + 5.$$

$$\frac{dD^2}{dt} = 0 \implies t = \frac{2}{7}.$$

If a student attempts to use Lagrange multiplier method, then 2 points for correctly specify the Lagrangian, 3 points for finding the solution, and 1 point for arguing that the obtained solution is the global minimum.

Q6 Answer:**Step 1: Find the Critical Points**

To find the critical points, we first compute the partial derivatives of $f(x, y)$ and set them equal to zero.

$$\frac{\partial f}{\partial x} = 3x^2 - 3y, \quad \frac{\partial f}{\partial y} = 3y^2 - 3x.$$

Set the partial derivatives to zero:

$$\begin{aligned} 3x^2 - 3y = 0 &\Rightarrow y = x^2, \\ 3y^2 - 3x = 0 &\Rightarrow x = y^2. \quad (2 \text{ points}) \end{aligned}$$

Substitute $y = x^2$ into the second equation:

$$x = (x^2)^2 \Rightarrow x = x^4 \Rightarrow x(x^3 - 1) = 0.$$

This gives $x = 0$ or $x = 1$.

- For $x = 0$, $y = 0^2 = 0$. So, one critical point is $(0, 0)$.

- For $x = 1$, $y = 1^2 = 1$. So, another critical point is $(1, 1)$. (2 points)

Step 2: Classify the Critical Points

Compute the second partial derivatives:

$$\frac{\partial^2 f}{\partial x^2} = 6x, \quad \frac{\partial^2 f}{\partial y^2} = 6y, \quad \frac{\partial^2 f}{\partial x \partial y} = -3.$$

The Hessian determinant D is:

$$D = \left(\frac{\partial^2 f}{\partial x^2} \right) \left(\frac{\partial^2 f}{\partial y^2} \right) - \left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 = (6x)(6y) - (-3)^2 = 36xy - 9. \quad (2 \text{ points})$$

Critical Point $(0, 0)$: $D = 36(0)(0) - 9 = -9 < 0$. Since $D < 0$, $(0, 0)$ is a saddle point.

Critical Point $(1, 1)$: $D = 36(1)(1) - 9 = 27 > 0$, and $\frac{\partial^2 f}{\partial x^2} = 6 > 0$.

Since $D > 0$ and $\frac{\partial^2 f}{\partial x^2} > 0$, $(1, 1)$ is a local minimum.

Final Answer

The function $f(x, y) = x^3 + y^3 - 3xy$ has:

- A saddle point at $(0, 0)$. (1 points)
- A local minimum at $(1, 1)$. (1 points)

Q7 Answer:

(i)

$$\begin{aligned} & \iint_R (3 - x - y) \, dA \quad (3 \text{ points}) \\ &= \int_{-1}^2 \int_{x^2}^{x+2} (3 - x - y) \, dy \, dx \\ &= \left[3y - xy - \frac{1}{2}y^2 \right]_{y=x^2}^{y=x+2} \\ &= \int_{-1}^2 3(x+2) - x(x+2) - \frac{1}{2}(x+2)^2 - \left[3x^2 - x^3 - \frac{1}{2}x^4 \right] \, dx \\ &= \int_{-1}^2 \frac{1}{2}x^4 + x^3 - \frac{9}{2}x^2 - x + 4 \, dx \quad (1 \text{ points}) \\ &= \left[\frac{x^5}{10} + \frac{x^4}{4} - \frac{3x^3}{2} - \frac{x^2}{2} + 4x \right]_{-1}^2 \\ &= \frac{81}{20} = 4.05 \quad (2 \text{ points}) \end{aligned}$$

(ii)

$$\begin{aligned} & \int_0^3 \int_{\sqrt{x/3}}^1 e^{y^3} \, dy \, dx \\ &= \int_0^1 \int_0^{3y^2} e^{y^3} \, dx \, dy \quad (4 \text{ points}) \\ &= \int_0^1 3y^2 e^{y^3} \, dy = \left[e^{y^3} \right]_0^1 = e - 1 \quad (2 \text{ points}) \end{aligned}$$

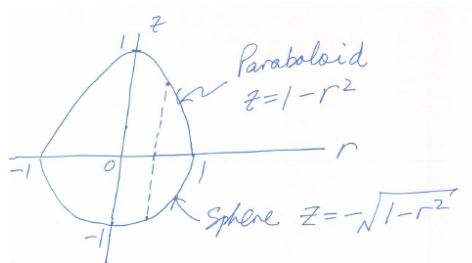


Figure 1: Q8: Integration region G (shaded)

Q8 Answer:

(a)

The paraboloid is given by: $z = 1 - r^2$. The sphere is given by: $z = -\sqrt{1 - r^2}$

Projected view on a plane of constant θ : $z = 1 - r^2$ The volume V is:

$$V = \int_0^{2\pi} \int_0^1 \int_{-\sqrt{1-r^2}}^{1-r^2} dz r dr d\theta \quad (4 \text{ points})$$

(b)

$$\begin{aligned} V &= \int_0^{2\pi} \int_0^1 \left[(1 - r^2) - (-\sqrt{1 - r^2}) \right] r dr d\theta = \int_0^{2\pi} \int_0^1 (-r^2 + \sqrt{1 - r^2}) r dr d\theta \\ &= \pi \left[\int_0^1 (1 - u) du + \int_0^1 (\sqrt{1 - u}) du \right] = \frac{7}{6}\pi \quad (4 \text{ points}) \end{aligned}$$

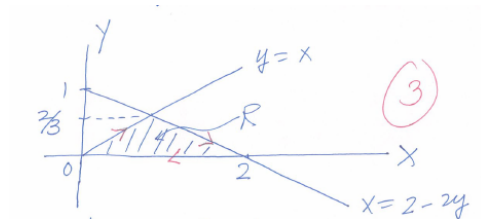


Figure 2: Q9: Integration region R (shaded)

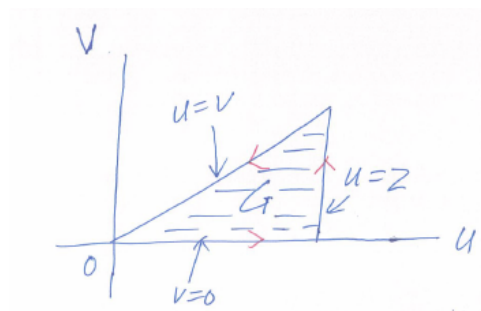


Figure 3: Integration region G (shaded)

Q9 Answer:

(a): $I = \int_0^{2/3} \int_y^{2-2y} (x + ay) e^{y-x} dx dy$

The integration region is R (shaded). (3 points)

(b): Transformation: $u = x + 2y$, $v = x - y$ Solving gives: $x = \frac{u+2v}{3}$, $y = \frac{u-v}{3}$ Reimage G of R : $y = x \Rightarrow v = 0$ $x = 2 - 2y \Rightarrow u = 2$ $y = 0 \Rightarrow u = v$

The Jacobian J is: $J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{3} & \frac{2}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{vmatrix} = -\frac{1}{3}$ The integral becomes:

$$I = \iint_G u e^{-v} |J| du dv = \frac{1}{3} \iint_G u e^{-v} du dv = \frac{1}{3} \int_0^2 \int_{v=0}^u u e^{-v} dv du \quad (5 \text{ points})$$

Q10 Answer:

(a) For $\mathbf{F} = P\mathbf{i} + Q\mathbf{j} + R\mathbf{k}$ to be a gradient field, the following must hold:

$$\begin{aligned}\frac{\partial P}{\partial y} &= \frac{\partial Q}{\partial x} \\ \frac{\partial P}{\partial z} &= \frac{\partial R}{\partial x} \\ \frac{\partial Q}{\partial z} &= \frac{\partial R}{\partial y}\end{aligned}$$

Condition 1:

$$\frac{\partial P}{\partial y} = 2y, \quad \frac{\partial Q}{\partial x} = by$$

Setting them equal:

$$2y = by \implies b = 2 \quad (2 \text{ points})$$

Condition 2:

$$\frac{\partial P}{\partial z} = 2cx, \quad \frac{\partial R}{\partial x} = 2cx$$

This condition is satisfied for all c .

Condition 3:

$$\frac{\partial Q}{\partial z} = cy, \quad \frac{\partial R}{\partial y} = 2y$$

Setting them equal:

$$cy = 2y \implies c = 2 \quad (2 \text{ points})$$

Final Answer: $\boxed{b = 2}$ and $\boxed{c = 2}$.

(b) Step 1: $\frac{\partial \phi}{\partial x} = y^2 + 4xz$. Integrate (1) with respect to x :

$$\phi(x, y, z) = \int (y^2 + 4xz) dx = xy^2 + 2x^2z + C(y, z) \quad (1 \text{ points})$$

Step 2: Differentiate ϕ with respect to y and compare with $\frac{\partial \phi}{\partial y} = 2xy + 2yz$:

$$\frac{\partial \phi}{\partial y} = 2xy + \frac{\partial C(y, z)}{\partial y} = 2xy + 2yz \implies \frac{\partial C(y, z)}{\partial y} = 2yz$$

Integrate with respect to y :

$$C(y, z) = y^2z + D(z) \quad (1 \text{ points})$$

Step 3: Substitute $C(y, z)$ back into ϕ :

$$\phi(x, y, z) = xy^2 + 2x^2z + y^2z + D(z) \quad (1 \text{ points})$$

Step 4: Differentiate ϕ with respect to z and compare with $\frac{\partial \phi}{\partial z} = y^2 + 2x^2$:

$$\frac{\partial \phi}{\partial z} = 2x^2 + y^2 + D'(z) = y^2 + 2x^2 \implies D'(z) = 0$$

Integrate with respect to z :

$$D(z) = K$$

Final Answer: The potential function is $\boxed{\phi(x, y, z) = xy^2 + 2x^2z + y^2z + K}$ (1 points).
where K is an arbitrary constant.

(c) Since F is a gradient field with potential function ϕ , the work done by F along any path from point A to point B is given by:

$$W = \phi(B) - \phi(A)$$

1. At $A = (0, 0, 0)$:

$$\phi(0, 0, 0) = 0 + 0 + 0 + K = K$$

2. At $B = (1, 2, 2)$:

$$\phi(1, 2, 2) = (1)(2)^2 + 2(1)^2(2) + (2)^2(2) + K = 4 + 4 + 8 + K = 16 + K$$

Work Done:

$$W = \phi(B) - \phi(A) = (16 + K) - K = 16$$

Final Answer: The work done by F is $\boxed{16}$ (4 points).

Q11 Answer:

The flux is given by $\oint_C \mathbf{F} \cdot \mathbf{n} ds$, where $\mathbf{n}$ is the outward unit normal vector.

1. Right Semicircular Path Parameterization:

$$\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t \rangle, \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

Outward normal:

$$\mathbf{n} = \langle \cos t, \sin t \rangle$$

Field evaluated on path:

$$\mathbf{F} = \langle 2 \cos t, 2 \sin t \rangle$$

Flux integral:

$$\int_{C_1} \mathbf{F} \cdot \mathbf{n} ds = \int_{-\pi/2}^{\pi/2} (2 \cos^2 t + 2 \sin^2 t) \cdot 2 dt = 4 \int_{-\pi/2}^{\pi/2} dt = \boxed{4\pi} \quad (2 \text{ points})$$

2. Line Segment Parameterization:

$$\mathbf{r}(t) = \langle 0, t \rangle, \quad -2 \leq t \leq 2$$

Outward normal:

$$\mathbf{n} = \langle -1, 0 \rangle$$

Flux integral:

$$\int_{C_2} \mathbf{F} \cdot \mathbf{n} ds = \int_{-2}^2 -0 dt = \boxed{0} \quad (2 \text{ points})$$

Total Flux

$$\text{Total Flux} = 4\pi + 0 = \boxed{4\pi}$$

(b) The circulation is given by $\oint_C \mathbf{F} \cdot d\mathbf{r}$.

1. Right Semicircular Path Parameterization:

$$\mathbf{r}(t) = \langle 2 \cos(-t), 2 \sin(-t) \rangle, \quad -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

Tangent vector:

$$\mathbf{r}'(t) = \langle -2 \sin t, -2 \cos t \rangle$$

Field evaluated on path:

$$\mathbf{F} = \langle 2 \sin t, 2 \cos t \rangle$$

Circulation integral:

$$\int_{C_1} \mathbf{F} \cdot d\mathbf{r} = \int_{-\pi/2}^{\pi/2} (-4 \sin^2 t - 4 \cos^2 t) dt = -4 \int_{-\pi/2}^{\pi/2} dt = \boxed{-4\pi} \quad (2 \text{ points})$$

2. Line Segment Parameterization:

$$\mathbf{r}(t) = \langle 0, t \rangle, \quad -2 \leq t \leq 2$$

Tangent vector:

$$\mathbf{r}'(t) = \langle 0, 1 \rangle$$

Field evaluated on path:

$$\mathbf{F} = \langle t, 0 \rangle$$

Circulation integral:

$$\int_{C_2} \mathbf{F} \cdot d\mathbf{r} = \int_{-2}^2 0 dt = \boxed{0} \quad (2 \text{ points})$$

Total Circulation

$$\text{Total Circulation} = -4\pi + 0 = \boxed{-4\pi}$$

Q12 Answer:

The outward flux of $\mathbf{F}$ through the surface S of the cube is given by the divergence theorem:

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{F}) dV, \quad (2 \text{ points})$$

where V is the volume of the cube.

First, compute the divergence of $\mathbf{F}$:

$$\nabla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x^2y) + \frac{\partial}{\partial y}(y^2z) + \frac{\partial}{\partial z}(z^2x) = 2xy + 2yz + 2zx. \quad (2 \text{ points})$$

Now, evaluate the triple integral over the cube $V = [0, 1] \times [0, 1] \times [0, 1]$:

$$\iiint_V (2xy + 2yz + 2zx) dV = \int_0^1 \int_0^1 \int_0^1 (2xy + 2yz + 2zx) dz dy dx. \quad (1 \text{ points})$$

Compute the innermost integral with respect to z :

$$\int_0^1 (2xy + 2yz + 2zx) dz = 2xy \cdot 1 + 2y \cdot \frac{1}{2} + 2x \cdot \frac{1}{2} = 2xy + y + x. \quad (1 \text{ points})$$

Next, integrate with respect to y :

$$\int_0^1 (2xy + y + x) dy = 2x \cdot \frac{1}{2} + \frac{1}{2} + x \cdot 1 = x + \frac{1}{2} + x = 2x + \frac{1}{2}. \quad (1 \text{ points})$$

Finally, integrate with respect to x :

$$\int_0^1 \left(2x + \frac{1}{2}\right) dx = 2 \cdot \frac{1}{2} + \frac{1}{2} \cdot 1 = 1 + \frac{1}{2} = \frac{3}{2}. \quad (1 \text{ points})$$

Q13 Answer:

By Stokes' Theorem, the surface integral of the curl of $\mathbf{F}$ over S is equal to the line integral of $\mathbf{F}$ over the boundary curve C :

$$\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, d\sigma = \oint_C \mathbf{F} \cdot d\mathbf{r}. \quad (2 \text{ points})$$

Step 1: Parameterize the Boundary Curve C

The boundary C is the circle where the paraboloid intersects the xy -plane ($z = 0$): $4 - x^2 - y^2 = 0 \implies x^2 + y^2 = 4$. We can parameterize C using polar coordinates: $\mathbf{r}(t) = \langle 2 \cos t, 2 \sin t, 0 \rangle$, $0 \leq t \leq 2\pi$. (2 points)

Step 2: Compute $\mathbf{F}$ on C

Substitute the parameterization into $\mathbf{F}$: $\mathbf{F}(\mathbf{r}(t)) = \langle (2 \sin t)^2, 0^2, (2 \cos t)^2 \rangle = \langle 4 \sin^2 t, 0, 4 \cos^2 t \rangle$. (1 points)

Step 3: Compute $d\mathbf{r}$

Differentiate $\mathbf{r}(t)$ with respect to t : $d\mathbf{r} = \langle -2 \sin t, 2 \cos t, 0 \rangle dt$. (1 points)

Step 4: Compute the Line Integral

The line integral becomes: $\oint_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \langle 4 \sin^2 t, 0, 4 \cos^2 t \rangle \cdot \langle -2 \sin t, 2 \cos t, 0 \rangle dt$.
Simplify the dot product: $= \int_0^{2\pi} (4 \sin^2 t)(-2 \sin t) + (0)(2 \cos t) + (4 \cos^2 t)(0) dt = \int_0^{2\pi} -8 \sin^3 t dt$. (1 points)

Step 5: Evaluate the Integral

The integral of $\sin^3 t$ over $[0, 2\pi]$ is zero because $\sin^3 t$ is an odd function and the interval is symmetric: $\int_0^{2\pi} \sin^3 t dt = 0$. Thus, $\oint_C \mathbf{F} \cdot d\mathbf{r} = -8 \times 0 = 0$. (1 points)

Final Answer

The surface integral is: $\boxed{0}$.